INCH-POUND

MIL-PRF-53013B (AT)
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28 October 2022
SUPERSEDING
MIL-PRF-53013B (AT)
01 February 1999

PERFORMANCE SPECIFICATION

REPAIR KIT, 6-INCH HOSELINE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a repair kit consisting of tools and materials necessary for permanent repair to the Inland Petroleum Distribution System (IPDS) 6-inch, lightweight, collapsible hose-line. The components are contained in a fitted storage chest.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4610

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COMMERCIAL ITEM DESCRIPTIONS

A-A-52598 - Couplings, Clamp, Pipe; with Bolts and Synthetic-Rubber Gaskets for Grooved-End Pipe

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Description. The hose-line repair kit (hereinafter called "kit") shall consist of the components described herein and be fitted in a storage chest.

3.2 Materials. Materials shall be selected by the contractor and shall be subject to all provisions of this specification. Materials selected shall meet or exceed the requirements specified herein and shall be sufficient to withstand the rigors of daily use in the applicable operating and storage environments to which the parts may be exposed, without suffering the effects of deterioration. Storage environment for the repair kit is - 40 °F to 160 °F and operating environment is -20 °F to 135 °F respectively with humidity of 0-100%.

3.2.1 Identification of materials and finishes. The contractor shall identify and document the specific materials, material finish, or treatment for use with components and sub-components.

3.2.2 Recovered materials. The offerer/contractor is encouraged to use recovered materials (see 6.5.2) to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

3.2.3 Deterioration prevention and control. The equipment shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration to which the materials are susceptible.

3.2.3.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion.

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3.3 Repair kit parts list. The components comprising the repair kit and their required quantities are listed in Table I.

TABLE I. 6-Inch hose-line repair kit parts.

Qty.	Description	Ref No.
3	Coupling, clamp, 6 inch nom. size	A-A-52598
6	Gasket, coupling, 6 inch nom. size	A-A-52598
2	Clamp, hose, 6 inch	(figure 1)
18	Hose nipple assembly, grooved end, three-segment, 6 inch	(figure 2)
6	Hose repair assembly, grooved end, three-segment, 6 inch	(figure 3)
6	Adapter assembly, hose, double grooved end, three-segment, 6 inch	(figure 4)
1	Set, wrench, socket, 12 point, 1/2 drive	(3.6.1)
1	Shears, 8 inch nom. size.	(3.6.2)
1	Knife, rubber cutting, non- conductive handle, 8.00 in. × 1.00 in.	(3.6.3)

3.4 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.5 Hose clamp. The hose clamp shall consist of a base, fastener, frame, crosshead, and a screw and screw handle (pictured) or other means for lowering the crosshead and compressing the hose (see figure 1).

3.5.1 Base. Means shall be provided for latching and pivoting the vertical frame.

3.5.2 Fastener. The fastener shall be self-actuating and affixed to the base in such a manner that it may move freely. Means shall be provided for quickly engaging and locking the movable side of the frame.

3.5.3 Frame. The frame shall be U-shaped. One side of the frame shall act as a hinge, mounted on the base, and the other shall have the capability of being securely engaged by the fastener. The inside of the frame shall have parallel tracks, free of burrs and fins, to act as guide for the crosshead. The crosshead shall slide over the entire length without binding. The frame shall have not less than 11.50 inch clearance between the legs of the “U” and shall provide not less than 6.50 inch clearance between the lower jaw and the crosshead when the frame is closed and the crosshead is in its uppermost position.

3.5.4 Crosshead. The top of the crosshead shall incorporate a method for quickly lowering the crosshead so that a hose can be compressed against the lower jaw and stop the flow of fluid through the hose without damaging the hose.

3.5.5 Handle. The handle shall move without binding in the screw and shall be equipped at each end with a means to prevent the handle from pulling out.

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3.6 Tools.

3.6.1 Set, wrench, socket. The socket set shall provide an assortment of sockets which will as a minimum, fit all the bolts provided with the hose clamps and assemblies. In addition, sockets shall be 12 point (double hexagon), 1/2 drive, regular length.

3.6.2 Shears. A pair of shears shall be provided for cutting IPDS 6-inch lightweight, collapsible hose line.

3.6.3 Knife. The rubber cutting knife shall have a non-conductive handle and a blade with at least 6 inches in length., suitable for cutting rubber hose. The handle shall be either impervious, or treated to minimize, impregnation by fuel and oil and designed to promote a positive gripping surface.

3.7 Storage chest. All components of the repair kit shall be fitted in a storage chest. The chest shall have a hinged cover, fastening devices, and handles. Bins, cradles, or blocking for components of the repair kit shall be provided to prevent damage during shipment or handling. A loading plan indicating the location of all items in the chest shall be provided. The loading plan shall be permanently attached to the inside of the cover.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. When a first article inspection is required, it shall be performed on one complete unit. The inspection shall include the examination 4.4 and the testing 4.5.

4.3 Conformance inspection. Conformance inspection shall include the examination 4.4.

4.4 Examination. Each repair kit shall be physically examined for compliance with the requirements specified in section 3. Any redesign or modification to comply with specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all examinations of performance, safety, human engineering, and dimensional requirements. Non-compliance with any specified requirement, or the presence of one or more defects shall constitute cause for rejection.

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4.5 Testing. The repair kit shall be tested by using the hose clamp (see 3.5) to clamp an IPDS 6-inch, lightweight, collapsible hose-line and repairing it as follows:

1. Pressurize a plugged test hose segment to 200 psi minimum with water.
2. Place the hose clamp around the hose approximately 3 feet from the plugged end of the test hose segment.
3. Apply sufficient pressure with the hose clamp to the test hose to stop water flow.
4. After the hose has been compressed to stop fluid flow, cut the hose half-way between the clamp and the hose end-plug.
5. Any continuing flow (after initial draining) greater than 1 cup per minute shall constitute failure of this test.
6. Reattach the cut hose ends with a grooved end hose repair assembly (see figure 3).
7. Disengage the hose clamp.
8. The hose shall be checked for leakage after being re-pressurized and at half-hour intervals for the next 3 hours.
9. There shall be no damage to the hose at point of hose clamp engagement. Leakage of the hose at the repair site, and any permanent deformation of the hose clamp or hose hardware shall constitute failure.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The kit is intended for use in making permanent repairs on the Inland Petroleum Distribution System (IPDS) 6-inch, lightweight, collapsible hose-line. The hose clamps should be used to crimp pressurized hoses on both sides of a damaged section and enable the repair crew to cut the damaged section out and install hose couplings or repair assemblies contained in the kit, as required, to effect a repair with minimum time and effort.

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6.2 Acquisition requirements. Acquisition documents will specify the following:

a. Title, number, and date of this specification
b. Issue of ASSIST to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2)
c. When first article equipment is not required or time frame for submission of first article (see 3.4)
d. Packaging requirements (see 5.1).

6.3 First article inspection. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 Data Requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractors maintenance and operation manuals to be furnished with each repair kit.

6.5 Definitions.

6.5.1 Dissimilar metals. For its purpose, this document considers metals dissimilar when two specimens in contact with each other promote accelerated galvanic corrosion (see 3.2.3.1).

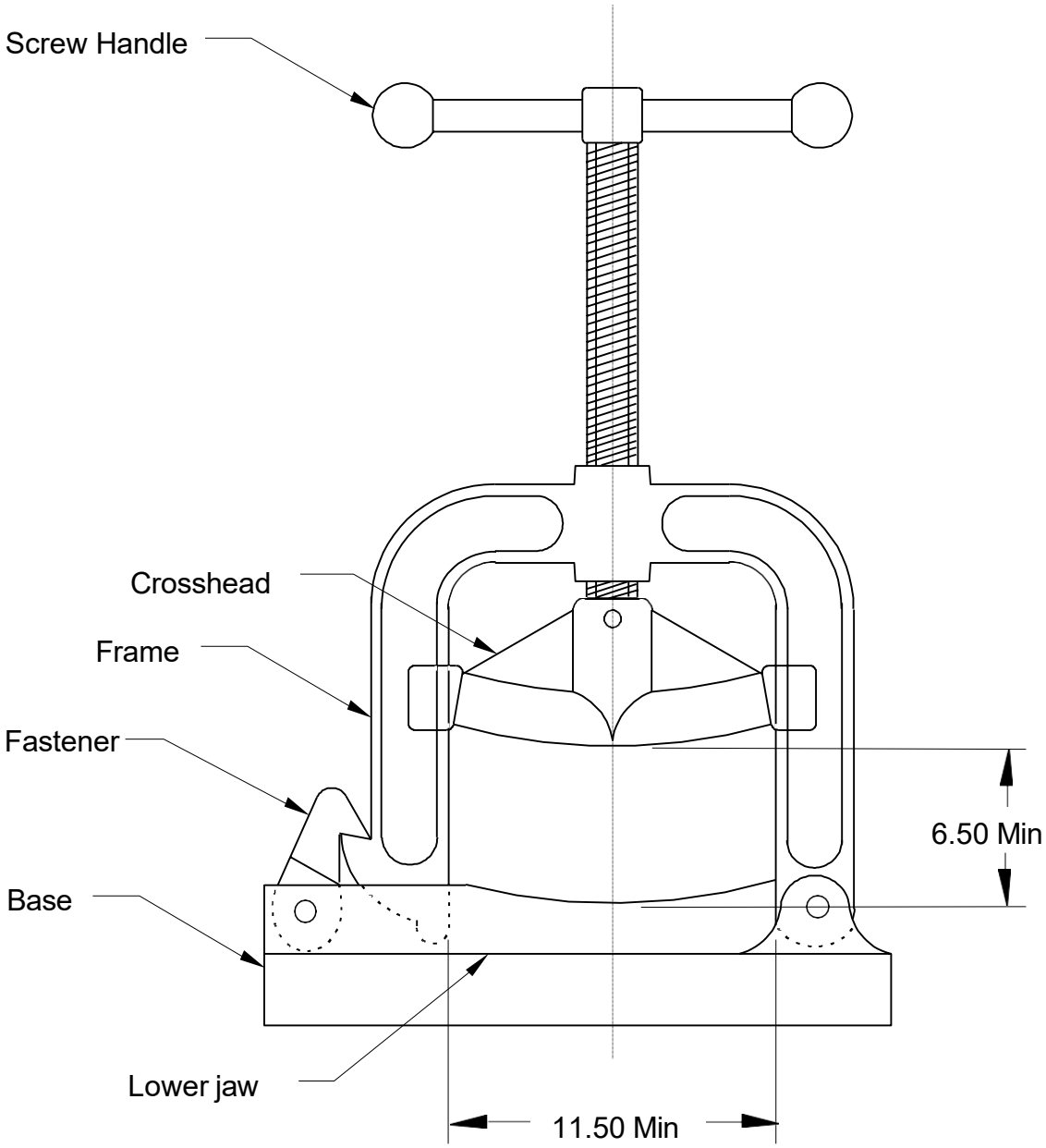
6.5.2 Recovered materials. For the purpose of this document, recovered materials are those materials that have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials (see 3.2.2).

6.6 Subject term (key word) listing.

Fuel, diesel
Fuel, aviation
Clamps, hose

6.7 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

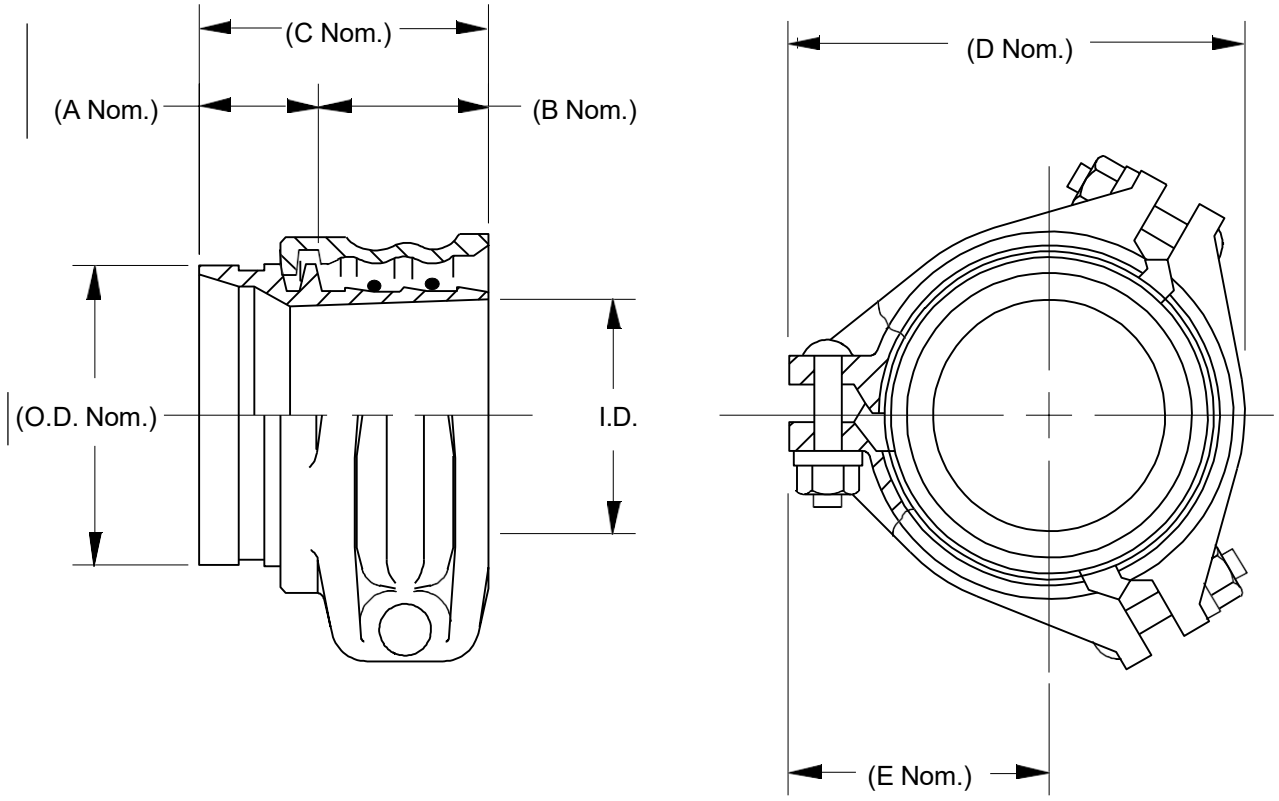
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NOTE: Dimensions are in inches.

FIGURE 1. Hose Clamp - Example.

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Nom. Size	Dimensions							Approx Wgt Lb*	Hose Wall Thickness		Hose ID
	A	B	I.D.	O.D.	E	D	C		Min	Max	
6.00	1.80	2.70	5.40	6.64	4.90	9.80	4.50	4.50	0.09	0.21	6.00

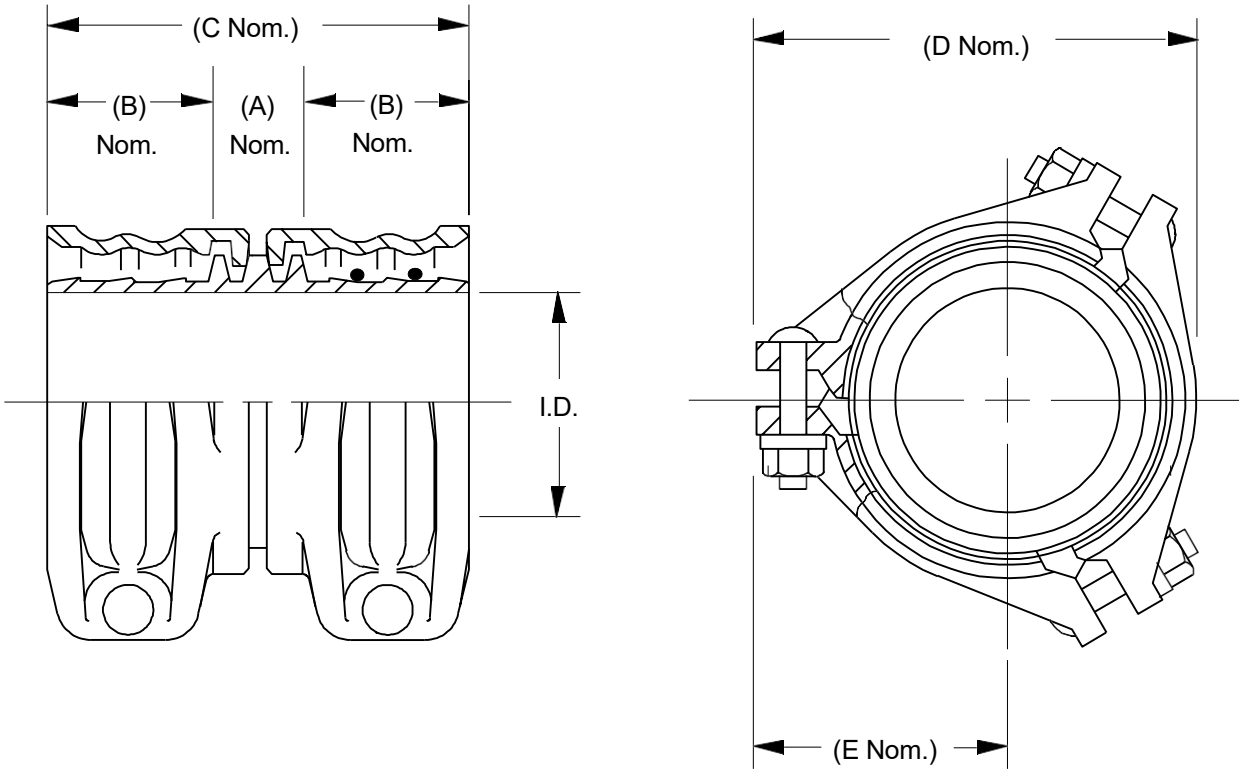
* Nipple and clamp assembly complete.

NOTES:

1. Dimensions are in inches.
2. Tolerances .XX ± .06.

FIGURE 2. Hose Nipple Assembly, Grooved End - Example.

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Nom. Size	Dimensions							Approx Wgt Lb*	Hose Wall Thickness		Hose I.D.
	A	B	I.D.	O.D.	E	D	C		Min	Max	
6.00	1.40	2.70	5.40	N/A	4.90	9.80	6.80	7.50	0.09	0.21	6.00

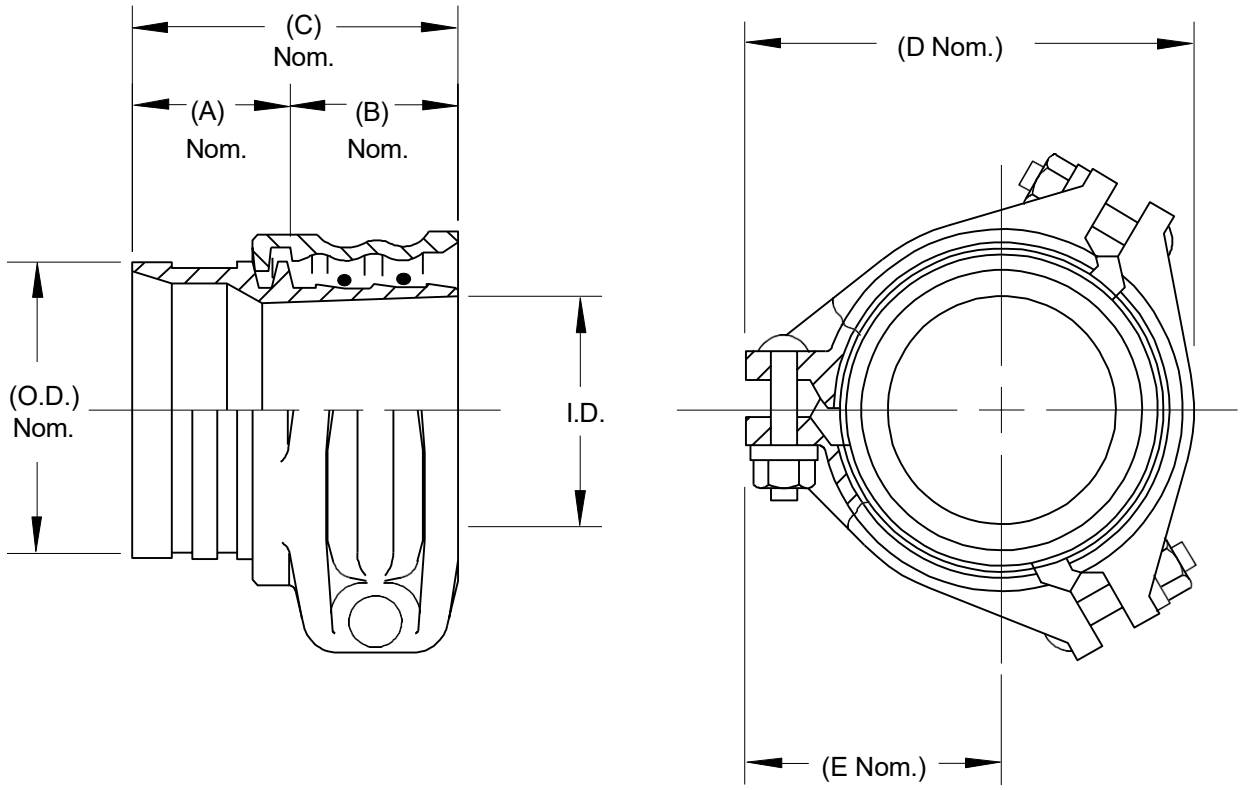
* Nipple and clamp assembly complete.

NOTES:

1. Dimensions are in inches.
2. Tolerances .XX ± .06.

FIGURE 3. Hose Repair Assembly, Grooved End - Example.

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Nom. Size Inches	Dimensions							Approx Wgt Lb*	Hose Wall Thickness		Hose I.D.
	A	B	I.D.	O.D.	E	D	C		Min	Max	
6.00	2.37	4.19	5.40	6.60	4.90	9.80	6.56	5.00	0.09	0.21	6.00

* Nipple and clamp assembly complete.

NOTE:

1. Dimensions are in inches.
2. Tolerances .XX ± .06.

FIGURE 4. Adapter Assembly, Hose, Double Grooved End - Example.

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CONCLUDING MATERIAL

Custodian:
Army - AT

Preparing activity:
Army - AT

Review activity:
DLA - CC

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